

Your Code

```
1 /*a) Read review id, product id, product anme, rating,reviiew text, re
2
3 we create a review class containing all the required fields;
4 class review{
5     String reviewid;
6     String productid;
7     String productname;
8     String reviewername;
9     int ratinng;
10    String reviewtext;
11    String reviewdate;
12    String vendorname;
13    review(String reviewid, String productid, String productname,String
14        this.reviewername=reviewername;
15        this.productid=productid;
16        this.productname=productname;
17        this.rating=rating;
18        this.reviewtext=reviewtext;
19        this.reviewdata=reviewdata;
20        this.vendorname=vendorname;
21    }
22 }
23
24
```

Input

```
4
good product
08-08-2026
abc electronics
```

Output

```
Error: Could not find or load
main class containing
Caused by:
java.lang.ClassNotFoundException:
containing
```

```
23
24
25 /*b) use string methods to cleanreviewer name, productnames and review
26
27 we can use string methods such as trim() and replaceAll()
28 this removes unnecessary spaces at the beginning or the end
29 this converts multiple spaces between words into a single space
30
31 reviewername=reviewername.trim().replaceAll("\\s+", " ");
32 productname=productname.trim().replaceAll("\\s+", " ");
33 reviewtext=reviewtext.trim().replaceAll("\\s+", " ");*/
34
35
36 /*c) store review records in arraylist
37
38 ArrayList<Review> reviews= new ArrayList<Review>();
39 reviews.add(review);
40
41 structure:
42     |
43     +---Review 1
44     +---REview 2
45     +---Review 3  */
46
47
48 /*d)remove duplicatereview ids using a hashset
49 a hashset stores unique review IDs. the add() method returns false if
50 code:
51 HashSet<String> reviewIds=new HashSet<>();
```

```

52 if(reviewIds.add(reviewId)){
53     reviews.add(review);
54 }
55 else{
56     System.out.println("Duplicate review ID");
57 } */
58
59
60 /*e) Maintain a hashmap mapping product ids with customer reviews
61
62 A hashmap is used to group reviews according to product id
63 HashMap<String,ArrayList<Review>>
64 Structure:
65     p101->review 1,review2
66     p102->review 3
67     p103->review 4
68 code:
69 HashMap<String,ArrayList<Review>> productreviews= new HashMap<String,
70 if(!productreviews.containsKey(productid)){
71 productreviews.put(productid, new ArrayList<Review>());
72 }
73 productreviews.get(productid).add(review); */
74
75 /* f) display reviews using iterator
76
77 an iterator is used to traverse the arraylist and display each review
78 code:
79 Iterator<review>=reviews.iterator();
80 while(it.hasNext()){

```

```

81     review r=it.next();
82     System.out.println(r.reviewid+" "+r.productname+" "+r.reviewername+
83 }*/
84
85 /* g)  implement generic collections
86
87 generic speify the type of data stored in a collection
88 example: ArrayList<review>
89           HashSet<String>
90           HashMap<String,ArrayList<review>>
91 Advantages:
92     1.provides type safety
93     2. Avoids unnecessary type casting
94     3. Makes the code easier  to understand */
95
96
97 /*h) generate product-wise review sumamries and explain the role of ja
98
99 the hashmap groups reviews by product id. we can then calculate the nu
100 code:
101 for(String: productid:productreviews.keySet()){
102     ArrayList<review>list=productreview.get(productid);
103     int total=0;
104     for(review r:list){
105         total=total+r.rating;
106
107     }
108     double average=(double) total/list.size();
109     System.out.println("product id:"+productid);

```

```
110     System.out.println("reviews:"+list.size());
111     System.out.println("Average Rating:"+average);
112 }*/
113
114 //WHOLE CODE
115 import java.util.*;
116 import java.lang.*;
117 class review{
118     String reviewid;
119     String productid;
120     String productname;
121     String reviewername;
122     String reviewtext;
123     int rating;
124     String reviewdate;
125     String vendorname;
126     review(String reviewid, String productid,String productname,String
127         this.reviewid=reviewid;
128         this.reviewername=reviewername;
129         this.productid=productid;
130         this.productname=productname;
131         this.rating=rating;
132         this.reviewdate=reviewdate;
133         this.vendorname=vendorname;
134         this.reviewtext=reviewtext;
135     }
136     public static void main(String[] args){
137         Scanner sc=new Scanner(System.in);
138         ArrayList<review> reviews=new ArrayList<review>();
```

```
139 HashSet<String>reviewids=new HashSet<String>();
140 HashMap<String,ArrayList<review>>productreviews=new HashMap<String,Arr
141 System.out.println("Enter the number of reviews:");
142 int n=sc.nextInt();
143 sc.nextLine();
144 for(int i=0;i<n;i++){
145     System.out.println("\nreview "+(i+1));
146     System.out.println("Review id:");
147     String reviewid=sc.nextLine().trim();
148     if(!reviewids.add(reviewid)){
149         System.out.println("duplicate review id");
150         continue;
151     }
152     System.out.println("product id:");
153     String productid=sc.nextLine().trim();
154     System.out.println("product anme:");
155     String productname=sc.nextLine().trim().replaceAll("\\s+", " ");
156     System.out.println("reviewer name:");
157     String reviewername=sc.nextLine().trim().replaceAll("\\s+", " ");
158     System.out.println("rating:");
159     int rating=sc.nextInt();
160     sc.nextLine();
161     System.out.println("reviewtext:");
162     String reviewtext=sc.nextLine().trim();
163     System.out.println("review date:");
164     String reviewdate=sc.nextLine().trim();
165     System.out.println("vendor name:");
166     String vendorname=sc.nextLine().trim().replaceAll("\\s+", " ");
167     review r=new review(reviewid,productid,productname,reviewername,re
```

```
168     reviews.add(r);
169     if(!productreviews.containsKey(productid)){
170         productreviews.put(productid,new ArrayList<review>());
171     }
172     productreviews.get(productid).add(r);
173 }
174 System.out.println("--ALL REVIEWS--");
175 Iterator<review>it=reviews.iterator();
176 while(it.hasNext()){
177     review r=it.next();
178     System.out.println("review id: "+r.reviewid);
179     System.out.println("product:"+r.productname);
180     System.out.println("reviewer: "+r.reviewername);
181     System.out.println("Rating:"+ r.rating);
182     System.out.println("review: "+r.reviewtext);
183     System.out.println("Date: "+r.reviewdate);
184     System.out.println("Vendor: "+r.vendorname);
185     System.out.println();
186 }
187 System.out.println("--SUMMARY");
188 for(String productid: productreviews.keySet()){
189     ArrayList<review>list=productreviews.get(productid);
190     int total=0;
191     for(review r:list){
192         total=total+r.rating;
193     }
194     double average=(double)total/list.size();
195     System.out.println("product id:"+productid);
196     System.out.println("product name:"+list.get(0).productname);
```